The Rarest Target Sets Passive Memory: An Exact Routing-Word Degree Law

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Abstract

We determine the exact passive memory required to route one fixed input subspace repeatedly among orthogonal spectral destinations. Let a routing word w of length L use target a exactly n_a times, and let every target and the input have dimension k. Among all square finite rational-inner transfer matrices, regular at the prescribed distinct boundary nodes and realizing the subspace routing constraints, we prove

$$d_{\min} = k\big(L - \min_{a\in\mathcal{A}(w)} n_a\big).$$

The lower bound is a zero budget: a compressed $k \times k$ minor vanishes to order at least k whenever the route points elsewhere, while its total zero budget cannot exceed McMillan degree. The matching construction is explicit. Node polynomials are normalized by scalar Fejér–Riesz factorization to form a rational-inner routing column; a degree-preserving lossless completion and tensor amplification produce a square network of the asserted degree.

Thus passive memory is governed by the rarest destination and, within this model, is invariant under every permutation of the routing word. Switch count need not even order the cost: for scalar channels, the cyclic word `AAAB` has two switches and degree three, whereas `ABAB` has four switches and degree two. The law recovers constant routing, a single switch, and fully distinct fan-out as exact limiting cases. It also yields the optimal integrated Wigner–Smith delay. A deterministic compiler and an independent verifier replay 1,024 random words, all 1,155 word classes through length seven modulo alphabet renaming, permutation tests, and clustered-node stress fixtures. The machine-readable certificate and the paper are linked to an immutable public artifact.

Keywords: rational inner matrix; McMillan degree; spectral routing; Fejér–Riesz factorization; paraunitary completion; passive memory; Wigner–Smith delay; routing word.

1 A word, not a path length

A lossless finite-dimensional discrete-time network is represented by a square rational matrix S analytic on $\mathbb{D}$ and unitary almost everywhere on $\mathbb{T}$. Its McMillan degree is the dimension of a minimal conservative realization. Fix distinct nodes $\zeta_1, \ldots, \zeta_L \in \mathbb{T}$, a k-plane $X \subset \mathbb{C}^N$, and mutually orthogonal k-planes $Y_a \subset \mathbb{C}^N$ indexed by a finite alphabet $\mathcal{A}$. A word $w = w_1 \cdots w_L \in \mathcal{A}^L$ prescribes

$$S(\zeta_i)X = Y_{w_i}, \qquad 1 \le i \le L. \tag{1}$$

Only the subspaces are prescribed; bases and phases are free. We minimize $\deg_{\mathrm{McM}} S$ over square finite rational-inner S regular at every node. Symbols not occurring in w are discarded, and

$$n_a := \#\{i : w_i = a\}, \qquad n_* := \min_{a\in\mathcal{A}(w)} n_a.$$

Previous exact results isolated the cost of one switch and of fully distinct orthogonal fan-out [7, 9]; the direct-sum sequel showed that maximal fan-out is geometrically generic [8]. Repeated targets create a different question: does memory count transitions, distinct targets, or interpolation constraints? The answer is a sharp occupancy statistic.

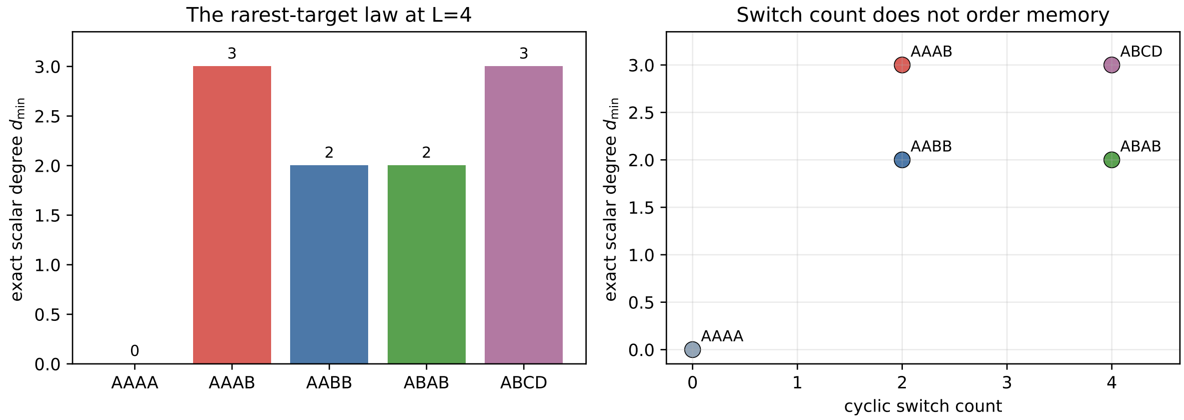

Figure 1: Exact scalar memory at length four. The law depends on the rarest occupancy, not monotonically on cyclic switch count. In particular, `AAAB` costs more than `ABAB`.

Theorem 1.1 (Rarest-target law)**.** *Under the hypotheses above,*

$$d_{\min}(w;k) = k(L - n_*). \tag{2}$$

The value is independent of the locations of the distinct nodes and of the order of the letters.

The order-independence in Theorem 1.1 concerns minimum state dimension, not every dynamical observable of a particular realization. It is already strong enough to separate memory from switch count.

2 A zero budget for compressed minors

Choose isometric frames X_0 for X and U_a for Y_a. For an interpolant S, define the scalar compressed minor

$$g_a(z) = \det(U_a^* S(z) X_0). \tag{3}$$

At a node carrying a, the matrix inside the determinant is unitary, up to the freely chosen input and output gauges; hence $g_a(\zeta_i) \neq 0$. At a node carrying $b \neq a$, orthogonality gives $U_a^* S(\zeta_i) X_0 = 0$. All k^2 entries therefore vanish there, and the determinant has a zero of order at least k.

We record the degree fact that converts these forced zeros into states.

Lemma 2.1 (Minor zero budget)**.** *Let S be a square finite rational-inner matrix of McMillan degree d, and let U, V be constant isometries with k columns. If $g(z) = \det(U^* S(z) V)$ is not identically zero, then the total multiplicity of its boundary zeros at regular points of S is at most d.*

Proof. Factor S into a constant unitary and d elementary rank-one Blaschke–Potapov factors, counting multiplicity [10]. On $\bigwedge^k \mathbb{C}^N$, g is the matrix coefficient $\langle \bigwedge^k U, (\bigwedge^k S) \bigwedge^k V \rangle$. Replacing

Table 1: Scalar examples. Cyclic switches count changes across all four adjacencies, including the last-to-first edge.

Word	L	occupancies	cyclic switches	$d_{\min}$	interpretation
`AAAA`	4	(4)	0	0	constant route
`AAAB`	4	(3,1)	2	3	rare excursion
`AABB`	4	(2,2)	2	2	two contiguous blocks
`ABAB`	4	(2,2)	4	2	alternating route
`ABCD`	4	(1,1,1,1)	4	3	full fan-out
`AABBCC`	6	(2,2,2)	3	4	balanced three-target route

one elementary factor by its scalar Blaschke coordinate changes $\bigwedge^k S$ affinely in that coordinate: an elementary rank-one factor has only one nonconstant invariant direction. After clearing the d scalar denominators, every matrix coefficient is a polynomial of degree at most d. A nonzero such numerator has at most d zeros, with boundary multiplicity included. Constant unitary factors and cancellations can only decrease this number. □

The exterior-power formulation is useful because it is insensitive to target-frame gauges. Replacing U_a by $U_a R_a$ and X_0 by $X_0 R_0$ multiplies g_a by the nonzero constant $\overline{\det R_a} \det R_0$; it cannot create or remove zeros. Nor can a pole be hidden at an interpolation node: regularity is explicitly part of the comparison class. These two details prevent a calibrated-frame or removable-singularity loophole in the lower bound.

Proposition 2.2 (Lower bound). *Every interpolant of* (1) *has degree at least* $k(L - n_*)$.

Proof. For a fixed used symbol a, the minor (3) is nonzero at its n_a own nodes and has a zero of order at least k at each of the other $L - n_a$ nodes. By Theorem 2.1, $\deg_{\mathrm{McM}} S \geq k(L - n_a)$. Taking an a of minimum occupancy proves the claim. □

This proof also explains why counting adjacent changes loses information. The obstruction is global: every destination monitors all nodes at which it is absent.

3 The optimal compiler

The lower bound is attained by a construction that first solves the scalar alphabet problem. Put $m = |\mathcal{A}(w)|$ and

$$p_a(z) = \prod_{i: w_i \neq a} (z - \zeta_i), \qquad \deg p_a = L - n_a \leq d := L - n_*. \tag{4}$$

The Laurent polynomial

$$R(z) = \sum_{a \in \mathcal{A}(w)} |p_a(z)|^2, \qquad z \in \mathbb{T}, \tag{5}$$

is strictly positive. Indeed, away from the interpolation nodes none of the p_a vanishes; at ζ_i, precisely the polynomial indexed by w_i is nonzero. Scalar Fejér–Riesz factorization gives an outer polynomial h of degree at most d, zero-free in $\mathbb{D}$, such that $|h|^2 = R$ on $\mathbb{T}$ [5]. Therefore

$$f(z) = \frac{1}{h(z)} (p_a(z))_{a \in \mathcal{A}(w)} \tag{6}$$

is an $m \times 1$ rational-inner column. At ζ_i all its coordinates except w_i vanish and the surviving coordinate has modulus one.

Rectangular rational-inner functions admit square lossless completions with the same realization state dimension [1, Theorem 2.1(ii)]; related polynomial spectral completions appear in [6]. Complete f to an $m \times m$ rational-inner V without increasing its degree. The scalar case of Theorem 2.2 forces $\deg_{\mathrm{McM}} V = d$, so no concealed cancellation is possible.

For rank k, take $V \otimes I_k$. Its McMillan degree is kd: each scalar elementary factor is repeated in k independent channels. Constant unitaries align the first input k-plane and the m coordinate target k-planes with X and Y_a; an identity block embeds the construction if $N > mk$. Constant operations change no degree. We have proved the matching upper bound and hence Theorem 1.1.

For completeness, the compiler can be stated as finite exact algebra before any numerical realization step:

1. delete unused symbols and count n_a;
2. form the node polynomials (4);

3. compute the positive trigonometric polynomial R in (5);
4. choose its outer Fejér–Riesz factor h and form f in (6);
5. apply a degree-preserving square lossless completion to f;
6. tensor by I_k and conjugate by constant input/output unitaries.

The first four steps already give a minimal scalar routing column. The fifth does not add states: in realization language it appends input and output channels around the same state matrix. The sixth repeats the scalar state space k times. This makes the upper proof constructive at the transfer, realization, and network-size levels.

Remark 3.1 (Algorithmic content). The proof is a compiler: construct the p_a, spectrally factor their summed energy, form f, complete losslessly, tensor, and align. No nonlinear search over network parameters is required.

4 Consequences and separations

Corollary 4.1 (Exact special cases). *The law gives: (i) constant routing, $d_{\min} = 0$; (ii) two distinct nodes routed to two targets, $d_{\min} = k$; (iii) L distinct targets, $d_{\min} = k(L-1)$; and (iv) balanced use of m targets when $m \mid L$, $d_{\min} = kL(1-1/m)$.*

Example 4.2 (More switches, less memory). For $k = 1$, `AAAB` has two cyclic switches, occupancies $(3, 1)$, and degree 3. The alternating word `ABAB` has four cyclic switches, occupancies $(2, 2)$, and degree 2. Thus switch count does not monotonically predict passive memory. Both examples use the same two target lines, so target span alone also fails to determine the answer.

Proposition 4.3 (Complete combinatorial quotient). *For fixed L and k, two routing words with orthogonal targets have the same minimum degree whenever their least nonzero occupancies agree. Every integer in*

$$k\lceil L/2\rceil,\ k(\lceil L/2\rceil + 1), \dots,\ k(L-1)$$

occurs among words using at least two targets, while the one-target word has degree zero.

Proof. The first statement is Theorem 1.1. With two used letters, the smaller occupancy can be any $r \in \{1, \dots, \lfloor L/2\rfloor\}$, giving degree $k(L-r)$. These are precisely the displayed values. More letters may repeat values but cannot fill a missing value because the degree is still $k(L-n_*)$. □

The discontinuity between one and two symbols is genuine. A word may use a second target only once; that single rare visit changes the minimum from zero to $k(L-1)$. Exact routing asks for a full k-plane zero at every other node, so rarity is expensive rather than cheap.

The theorem may be read as a law for a labelled walk on any graph whose vertices are represented by mutually orthogonal output planes: once the word of visited vertices is fixed, the minimum state dimension depends only on the least visit count. This is not a claim for nonorthogonal graph embeddings or edge-constrained internal dynamics.

For a differentiable unitary boundary value, use the Wigner–Smith matrix

$$Q(\theta) = -iS(e^{i\theta})^* \frac{d}{d\theta} S(e^{i\theta}). \tag{7}$$

The argument principle for a finite rational-inner matrix yields $\int_0^{2\pi} \operatorname{tr} Q(\theta) d\theta = 2\pi \deg_{\mathrm{McM}} S$. Hence:

Corollary 4.4 (Optimal integrated delay). *Among networks realizing the routing word,*

$$\min_S \int_0^{2\pi} \operatorname{tr} Q(\theta) d\theta = 2\pi k(L-n_*). \tag{8}$$

The equality is topological and global. It does not prescribe a unique pointwise delay profile, peak delay, or physical dwell-time distribution.

Table 2: Theorem-to-artifact map. Paths are relative to the public repository.

Claim	Executable witness
Occupancy law and compiler	`research/routing_word_memory_certificate.py`
Frozen campaign and residuals	`results/routing_word_memory/certificate.json`
Independent replay	`verification/verify_routing_word_memory.py`
Release-wide integrity gate	`verification/run_routing_word_memory_release_checks.py`
Switch-count separation	producer-generated `paper_routing_word_memory/figures/routing_word_memory.pdf`

4.1 Conservative-state interpretation

If $S(z) = D + zC(I - zA)^{-1}B$ is a minimal conservative realization, then $\deg_{\mathrm{McM}} S = \dim A$. The theorem therefore counts internal lossless state coordinates, not merely polynomial coefficients. In this representation the lower proof says that the same state space must supply $k(L - n_a)$ independent boundary-zero multiplicities for every target monitor a. The upper construction supplies exactly the largest of these demands and shares those states among the other monitors. This max-sharing mechanism is why the answer is

$$\max_a k(L - n_a) = k(L - n_*)$$

rather than the sum over targets.

The identity also separates architecture from calibration. Constant unitary pre- and post-networks can change the physical port basis at zero McMillan cost; only the frequency-dependent inner core contributes states. Conversely, allowing active gain or nonlossless cancellation would change the class and invalidate both the delay identity and the Potapov zero budget.

5 Executable certificate

The repository contains two deliberately separate programs. The producer constructs p_a, performs scalar spectral factorization by reciprocal-root selection, evaluates the inner column on a dense boundary grid, checks every node, freezes a JSON certificate, and creates Fig. 1. The verifier imports no producer code: it validates the schema and digest, recomputes the combinatorial law, and enforces numerical tolerances.

The frozen campaign contains 1,024 seeded random words of lengths two through nine; every one of the 1,155 restricted-growth word classes through length seven (alphabet renaming removed); 256 permutation checks; ten curated cases; and four `ABAB` fixtures with a closest prescribed angular gap decreasing from 10^{-1} to 10^{-4}. All pass. Across the full campaign the largest boundary column-isometry residual is 1.63×10^{-5}, the largest wrong-target leakage is 8.0×10^{-11}, and the smallest desired amplitude is 0.999996. For the exhaustive exact-grid subcampaign the corresponding isometry and leakage maxima are 1.1×10^{-13} and 4.4×10^{-15}.

The larger residual in clustered random fixtures is a conditioning warning for root-based floating-point factorization, not an exception to the exact identity. The certificate fails closed at stated tolerances; it is evidence that the constructive formulas were transcribed and replayed correctly, not a substitute for the proof.

The producer additionally stores the word, node angles, occupancies, predicted degree, spectral-factor residual, boundary isometry residual, desired amplitude, and wrong-coordinate leakage for each curated and stress case. A SHA-256 digest covers the canonical JSON payload before the final pass flag is appended. The independent verifier recomputes that digest and the degree formula from the stored words. This division catches both floating-point regression in the compiler and accidental mutation of the frozen evidence.

6 Scope, conditioning, and relation to prior work

Boundary matrix interpolation and McMillan-degree minimization have a deep general theory [2, 3], and modern subspace all-pass design can prescribe unitary data and group-delay blocks [4]. Fejér–Riesz factorization and lossless completion used here are classical. The contribution is their combination with the minor zero budget to expose a closed occupancy law for repeated orthogonal subspace routing. A targeted search of these literatures did not locate the formula (2); we therefore state novelty for this precise theorem, not for rational-inner interpolation in general.

Limitation 6.1 (Exact hypotheses). The theorem assumes distinct boundary nodes, exact subspace interpolation, mutually orthogonal equal-rank targets, and no prescribed frame inside a target. Repeated nodes require jet data. Nonorthogonal targets remove the forced-zero mechanism. Fully calibrated frames constrain phases discarded by (1). Active, nonrational, or nonlossless devices lie outside the comparison class.

6.1 Why the hypotheses are structural

Distinct nodes ensure that the polynomials (4) encode independent point constraints. If two nodes collide, value interpolation becomes inconsistent unless their labels agree; a meaningful limit with changing labels requires derivatives and hence a jet problem. Orthogonality is what makes the wrong-target compression identically zero. For merely separated targets it is small but nonzero, so the discrete zero count disappears even though approximate lower bounds may survive. Equal target ranks allow one tensor amplification; heterogeneous ranks would attach different determinant multiplicities to different absences and require a new packing argument.

Free target gauges are equally important. At a correct node the construction delivers the requested plane with a node-dependent phase. Fixing a complete unitary frame would turn those phases into data and can increase degree. The present theorem is consequently a statement about routing information, not waveform calibration.

Node positions do not alter exact degree, but they affect numerical conditioning and realization coefficients. Likewise, an arbitrarily small generic perturbation of a repeated target can turn repetitions into distinct direct-sum constraints and raise the exact degree, as the collision analysis in the direct-sum paper shows [8]. Robust approximate routing is therefore a different optimization problem.

Three extensions are natural. First, determine a stability law with explicit angular and subspace-error margins. Second, replace orthogonality by a graph of prescribed intersections and identify which zero budgets survive. Third, characterize local or peak delay among the degree-optimal completions. None is needed for the exact word law proved here.

7 Conclusion

Repeated spectral routing has a complete state-counting law. Each target asks the network to vanish, in a k-fold determinant sense, at every node assigned elsewhere. The least visited target therefore imposes the largest zero budget. Fejér–Riesz normalization realizes that budget without waste. The resulting identity $d_{\min} = k(L - n_*)$ is simultaneously a minimal-degree theorem, an explicit compiler, and a global delay law. Its most visible consequence is also the simplest: a route with more switches can require less memory.

Reproducibility. The canonical paper, source, frozen JSON certificate, independent verifier, and CI release gate are archived at the public repository.

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